Ashish Vashist

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Education

Carnegie Mellon University, Pittsburgh **Aug 2026 – Dec 2027**
M.S. in Machine Learning

Bharati Vidyapeeth College of Engineering, Pune **2020 – 2024**
Bachelor of Engineering, Computer Science & Engineering *CGPA 9.90/10*

Publications

- **Medmarks: A Comprehensive Open-Source LLM Benchmark Suite for Medical Tasks** — Benjamin Warner,..., **Ashish Vashist**, et al. *Under Submission - NeurIPS 2026* (Preprint)[Paper]
- **Where Do Vision-Language Models Fail? World Scale Analysis for Image Geolocalization** — Siddhant Bharadwaj, **Ashish Vashist**, Fahimul Aleem, Shruti Vyas *CVPR Workshop 2026 (Earthvision)* [Paper]
- **Beyond Accuracy: Evaluating Visual Grounding In Multimodal Medical Reasoning** — Anas Zafar, Leema Krishna Murli, **Ashish Vashist** *ICLR Workshop 2026 (CAO)* [Paper]
- **Multiscale Diagnostics of Visual Language Models** — Siddhant Bharadwaj, **Ashish Vashist**, Mohd Azfar, Rohit Kumar Salla, Divyansh Verma, Ramya Manasa Amancherla *ICCV Workshop 2025 (CV4DC)* [Paper]
- **FOD-S2R: A FOD Dataset for Sim2Real Transfer Learning based Object Detection — Ashish Vashist**, Qiranul Saadiyean, Suresh Sundaram, Chandra Sekhar Seelamantula [Paper]

Work Experience

Indian Institute of Science **August 2024 – Ongoing**
Pre-Doc Fellow *Bengaluru, Karnataka*

- Designed a 3D CAD replica of confined industrial environments and curated a first-of-its-kind synthetic dataset in Unreal Engine, incorporating variation in illumination spectra, FOVs, light reflections, sensor noise, object occlusion, and FOD size distributions.
- Conducted Sim-to-Real domain adaptation experiments and fine-tuned transformer-based object detection models, including Deformable DETR (DDQ), DINO, and Co-DETR, to improve object perception in severely constrained visibility settings.
- Developed HiDETR, a multiscale transformer-based hybrid detector, outperforming SOTA models with +4.7% improvement over YOLO and +1.6% over RT-DETR on VisDrone, COCO Val 2017 and FOD datasets.
- Currently researching Diffusion-based atmospheric turbulence mitigation, applying DDPM-driven restoration for long-range surveillance across multi-weather conditions (rain, haze, daylight variability).

Deep Forest Sciences **May 2023 – September 2023**
Machine Learning Intern *San Jose, California*

- Ported TextCNN and Deep Neural Network (DNN) architectures from TensorFlow to PyTorch, improving modularity, reproducibility, and ease of integration with modern research workflows.
- Enhanced DeepChem's open-source library module, fixing bugs, optimising data handling pipelines, and extending functionality.
- Added comprehensive unit tests, improving test coverage and ensuring long-term reliability of contributed modules.
- Improved developer documentation, examples, and API references, making the codebase easier for new contributors to adopt.

Contrails AI **May 2024 – July 2024**
Machine Learning Intern *Bengaluru, Karnataka*

- Researched and analysed modern DeepFake generation techniques across Logical Access (LA) and Text-to-Speech (TTS)–based methods, focusing on vulnerabilities exploited in multimedia systems.
- Designed and implemented a robust audio-spoofing detection pipeline for video segments, improving the integrity and authenticity verification of multimedia content.
- Assisted in identifying DeepFake videos circulated during the Indian General Elections 2024, collaborating with Times Now to support real-time media forensics and public misinformation detection.

Indian Institute of Information Technology **Feb 2022 – Sep 2022**
Full-Stack Intern *Prayagraj, Uttar Pradesh*

- Developed IIITA Alumni Analytics Project, collecting and processing alumni profiles from LinkedIn using Python-based web-scraping pipelines (BeautifulSoup, Selenium).
- Developed a full-stack GUI application for alumni database management using ElectronJS, Node.js, and MongoDB, enabling seamless search, filtering, and data visualisation.
- Implemented ML workflows, involving feature engineering, clustering, and supervised classification to segment alumni profiles and identify patterns in career progression and industry shifts.

Research Experience

Carnegie Mellon University *Pittsburgh, Pennsylvania*

Advisor: Prof. Min Xu — 3D Cryo-ET Subtomogram Classification

- Developed visual prompt tuning methods for 3D Cryo-ET subtomogram classification, achieving 63% accuracy and gaining hands-on experience with molecular-scale 3D data.
- Developed a visual prompt-tuning pipeline using Swin3D and equivariant 3D CNN encoders, enabling flexible modulation of volumetric transformer features.
- Designed a self-supervised learning framework combining multi-view rotational consistency, BYOL momentum updates, and SimCLR-style contrastive pretraining, achieving 63% accuracy on highly noisy cryo-ET volumes.
- Led a team of five undergraduates on few-shot 3D subtomogram classification, guiding literature reviews, ablations, and multi-stage training strategies.

University of Central Florida, USA *Orlando, Florida*

Advisor: Prof. Shruti Vyas — Geo-Localization with Fine-Tuned VLMs

- Evaluated Vision-Language Models on the task of predicting country-level geolocation from a single image across three benchmark datasets.
- Designed a systematic evaluation pipeline for VLM geolocation performance, comparing zero-shot accuracy, hallucination patterns, and robustness to image-quality variations.
- Applied LoRA-based parameter-efficient fine-tuning and Supervised Fine-Tuning (SFT) on curated geo-tagged datasets to improve country prediction accuracy and reduce regional confusion.

National University of Singapore *Singapore*

Advisor: Prof. Luo Wei — Geo-Spatial Temporal Modeling of COVID-19

- Analysed Facebook Human Movement data using GeoPandas, Shapely, and spatial-temporal processing to study mobility-driven COVID-19 trends across Singapore, Malaysia, and Indonesia.
- Built Python pipelines for OD matrices, flow aggregation, KDE heatmaps, and geospatial visualisations using Matplotlib, Folium, and Kepler.gl.
- Trained GCN, GAT, and GraphSAGE models with PyTorch Geometric to predict case growth using region-level mobility graphs and spatial embeddings.
- Conducted network analysis in Gephi to identify high-risk mobility corridors and super-connector regions.

Birla Institute of Technology *Goa*

Advisor: Prof. Swati Agarwal — Deep Learning Based Fall Detection

- Fall Detection Project using SisFall and UmaFall datasets, achieving 94.87% RNN accuracy, 96.33% Random Forest accuracy, and optimised thresholds across 12–16 IMU attributes.
- LSTM modelling on the combined dataset reaching 97% validation accuracy, supported by detailed confusion-matrix evaluations.
- Cross-dataset experiments using 67–72 feature configurations, consistently yielding 0.97–0.99 Random Forest accuracy and strong LSTM generalisation.
- Comprehensive fall-detection analysis using metadata-enhanced inputs and multiple feature variants with complete evaluation matrices across model setups.

Projects

Indic LLM Benchmarking & Reasoning Framework

- Co-led the creation of culturally grounded datasets across 20+ Indic languages, building scalable pipelines for data collection, cleaning, annotation, and validation with 80+ contributors using Python, HF datasets, and QA workflows.
- Designed evaluation benchmarks for cultural commonsense reasoning and temporal sequencing, and tested advanced reasoning LLMs under CoT, non-CoT, and structured-reasoning settings to measure cultural understanding.
- Collaborated with researchers from Hugging Face, Cohere AI, and an Alan Turing Fellow to develop a unified Indic LLM benchmarking suite, enabling systematic cross-model evaluation of multilingual and culturally grounded LLMs.
- Under preparation for EMNLP 2026

Contrastive Learning & Medical VLMs for Chest X-Ray Analysis

- Benchmarked MoCo v2, SimCLR, BYOL, and supervised contrastive learning on large-scale chest X-ray datasets to evaluate representation quality under limited-label regimes.
- Compared generalist (LLaVA, BLIP-2) and medical VLMs (BioViL, MedCLIP, CheXzero) using CoT and non-CoT prompting for diagnostic prediction and localisation.
- Fine-tuned SAMv3 for thoracic-region segmentation to enhance downstream classification and detection performance.

Technical Skills

Languages: Python, C/C++, SQL, HTML, CSS, JavaScript, MATLAB, LaTeX
Frameworks & Library: PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, FastAPI, Django, Keras, Streamlit, OpenCV, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, SQLAlchemy
Environment/Tools: Docker, Git/GitHub, Linux, CUDA, Anaconda, Google Colab, Jupyter, Tableau, MLflow, Weights & Biases, Unreal Engine, Tableau, PowerBI

Leadership / Extracurricular

CORD.ai — **2023 – Present**
Founder

- Founded CORD.ai , a non-profit deep learning research community that expanded to 800+ members across 12 countries, including contributors from IITs, CMU, Stanford, MIT, Microsoft, and Columbia.
- Led and organised multiple research subgroups in NLP, CV, and RL, enabling structured mentorship, collaborative project development, and cross-institution research engagement.
- Built a global learning ecosystem by hosting an AI symposium, running mentoring circles, and creating opportunities for students from smaller colleges to collaborate with international researchers and PhD mentors.

ETAAV — **August 2025**
Organizing Committee

- Served on the organising committee of the International Conference on Emerging Technology in Autonomous Aerial Vehicles (ETAAV), contributing to overall event coordination.
- Managed paper presentation scheduling, prepared slide decks, conducted plagiarism checks, and supported the Q&A and session-moderation team.

Community Service — **April 2023 – July 2023**
Tech Team Member

- Conducted multiple computer literacy and technology workshops for underserved students through local community initiatives.
- Taught and mentored HIV +ve children in collaboration with the Janseva Foundation, providing foundational digital education and academic support.

Achievements

- Selected for IIT Roorkee SPARK Internship Program (2022) out of 14,000+ students.
- Rank 1 in Smart India Internal Hackathon - 2022.
- Accepted to attend IIIT Hyderabad's CVIT, NYU AI Summer School, and ISI AI Winter School.